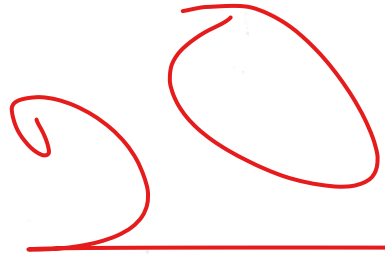


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Sec: 8



Here,

~~a is float [2][3]~~
~~Float has width 8~~

Here,

For a:

→ a is a float with dimension [2][3]

→ width, 8.

→ Total width of a = $2 \times 3 \times 8$
= 48

→ ~~offset~~ offset of a ~~is~~ starts with 0.

For b:

→ b is int type

→ width of int is 4

→ offset of b is after a, so it will be 48.

For P:

→ The record contains a and b

→ Total width of P is $= 48 + 4$
 $= 52$

→ The offset for P starts at 0.

Answer-2

Global scope:

No variable

Main function:

→ x : float

→ v : float

→ m : int [10] [20]

→ argcount : int

→ argvalue : char

→ t : float

→ n : int

→ for loop scope:

→ j : int

→ v : int

→ while loop scope:

→ count : int

→ n : float

→ m : float

Hash table:

① Global scope:

→ NO variable

② Main function:

(key) x → (value) float

(key) v → (value) float (outer scope, inner v)

(key) m → (value) int [10][20]

(key) orgCount → (value) int

(key) orgvalue → (value) char

(key) t → (value) float

(key) n → (value) int (outer scope, inner n)

③ For loop scope:

(key) i $\rightarrow$ (value) int

(key) s $\rightarrow$ (value) float

(key) c $\rightarrow$ (value) int

④ Inner for loop scope:

(key) j $\rightarrow$ (value) int

(key) v $\rightarrow$ (value) int [inner scope, outer v shadow]

⑤ while loop scope:

(key) Count $\rightarrow$ (value) int

(key) x $\rightarrow$ (value) float [inner scope, shadow ~~outer~~ outer x]

(key) m $\rightarrow$ (value) float (variable, not (x))

Answer-3

(a) Here are the three address code,

$$t_1 = a + a$$

$$t_2 = t_1 + t_1$$

$$t_3 = t_2 - a$$

$$t_4 = t_3 + a$$

$$t_5 = t_4 - t_2$$

b) For quadruples,

operator	arg1	arg2	result
+	a	a	t_1
+	t_1	t_1	t_2
-	t_2	a	t_3
+	t_3	a	t_4
-	t_4	t_2	t_5

For triples,

	Operator	Arg 1	Arg 2
0.	+	a	a
1.	+	(0)	(0)
2.	-	(1)	a
3.	+	(2)	a
4.	-	(3)	(1)

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